

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech Examination (CE)

December 2011

CE209: Design & Analysis of Algorithm

Date: 09.12.2011, Friday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

- | | | |
|-----|---|---|
| (a) | What will be the best case and worst case for Insertion-sort algorithm? | 2 |
| (b) | What is the smallest value of N such that an algorithm whose running time is $100n^2$ is runs faster than an algorithm whose running time is 2^n on the same machine? | 2 |
| (c) | What is meant by greedy Algorithm? What is the application of it? | 2 |
| (d) | Explain Optimal substructure Property with the help of an example. | 2 |
| (e) | Write an algorithm to find the minimum element in an array A storing n integers. What is the running time of this algorithm? | 3 |

Q-2

- | | | |
|-----|---|---|
| (a) | Consider the recurrence
$T(n) = 4T(n/2) + n^2$ When n is power of 2, $n \geq 2$.
Solve the above recurrence. | 6 |
| (b) | What are the difference between divide & conquer technique and dynamic programming? | 3 |
| (c) | Illustrate the operation of Merge Sort on the array $A = \{5, 13, 2, 25, 7, 17, 20, 8, 4\}$ | 3 |

OR

Q-2

- | | | |
|-----|--|---|
| (a) | Define Time Complexity. Explain with the help of a suitable example Best case, Worst Case and Average Case Time Complexity of an algorithm. | 4 |
| (b) | Differentiate between decision problem and optimization problem. | 4 |
| (c) | Given three matrixes $X_{3 \times 7}$, $Y_{7 \times 4}$, $Z_{4 \times 3}$. Find out optimal parenthesization using matrix chain multiplication algorithm. | 4 |

Q-3

- | | | |
|------------------|---|----|
| Attempt any Two. | | 12 |
| (a) | Let $S = \{a, b, c, d, e, f, g\}$ be a collection of objects with benefit-weight values as follows :
$a : (12, 4)$, $b : (10, 6)$, $c : (8, 5)$, $d : (11, 7)$, $e : (14, 3)$, $f : (7, 1)$, $g : (9, 6)$. What is an optimal solution to fractional knapsack problem using greedy algorithm assuming the sack can hold objects with total weight 18? | |
| (b) | What is the use of Recurrence Tree? Solve the given recurrence using recurrence Tree method. $T(n) = 3T(n/4) + n$ | |
| (c) | Sort the following list in increasing order of numbers using quick sort. Choose first element as a pivot element.
9, 94, 45, 47, 28, 98, 65, 42, 78, 4, 11, 88, 6 | |

SECTION-II

Q-4

- (a) What is Spanning tree? Give the Generic algorithm to find Spanning Tree for a given graph. 2
- (b) Arrange following functions in increasing order of their running time: 2
 $\log_2 n, 2^n, n^3, n^2 \log_2 n$
- (c) What do you mean by polynomial time complexity and exponential complexity? Which one is higher? 2
- (d) Write a recursive algorithm to compute the nth Fibonacci number. What is the running time of this algorithm? 2
- (e) What is an algorithm? List out the criteria that all algorithms should satisfy. 3

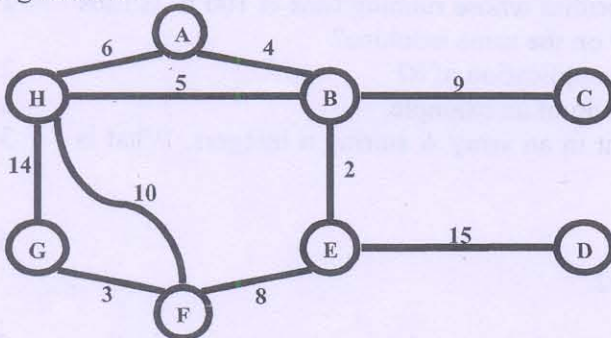
Q-5

- (a) What is Asymptotic Notation? Explain O, Ω , θ notation with a suitable example. 4
- (b) Write an algorithm for depth first search of a graph and explain with an example 4
- (c) Write an algorithm for naïve string matching. What is preprocessing and matching time for naïve string matching algorithm? 4

OR

Q-5

- (a) Which edges form the minimum spanning tree (MST) of the below graph using kruskal algorithm? 5



- (b) Define P, NP and NP-Complete problems. 3
- (c) Define Binary Search. Explain How to solve binary search problem using divide and conquer technique. 4

Q-6

- Attempt Any Two. 12
- (a) Given two sequences of characters: P=<100101101101> & Q=<01110>. Obtain the longest common subsequence.
- (b) Generate at least 2-solutions for 4-Queen's problem using backtracking. Draw a pruned state space tree for the problem.
- (c) Solve the following task assignment problem instance for minimization using branch and bound method.

	I	II	III	IV
A	4	7	3	12
B	2	6	1	8
C	3	9	4	11
D	12	3	18	2
